

**Life cycle inventories of global metal and mineral supply chains: a comprehensive data
review, analysis and processing**

Supporting Information (SI) document n°1

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1. Introduction

This supporting information (SI) document n°1 provides further details to the article regarding:

- The LCI data compilation, analysis and selection process (section 2).
- The Data verification and validation process (section 3).
- The rationale for defining the multifunctionality-solving approach (section 4).
- The background modelling rules, as considered in this study (section 5).
- The set of parameters and associated data sources and assumptions considered in the market shares estimation approach (section 6).
- Additional information on the LCI datasets selected in this study (section 7).

2. LCI data compilation, analysis and selection

2.1. Overall workflow

The literature review and LCI data analysis were carried out following a stepwise process, starting from literature screening and review of LCA studies (Figure S1). As a second step, LCI data identified in the LCA studies were compiled, for subsequent data quality and representativeness assessment. Based on this assessment and additional criteria (e.g. related to the market relevance of the production route), a selection of the most relevant LCI datasets for modelling the global market was performed.

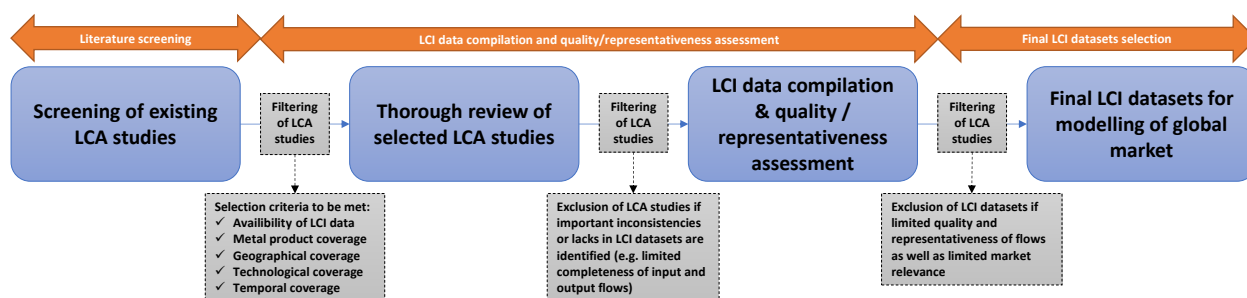


Figure S1: Literature review and LCI compilation, analysis and selection workflow

After screening and review of LCA studies, all identified LCI data were compiled under a harmonised format. This step aimed to gather all data as directly originating from the LCA studies, so as to prevent any data manipulation that could hinder the data quality/representativeness analysis. Along with the data (i.e. flow names, values, unit), additional information was reported to support the implementation of the Pedigree approach.

This data compilation step and quality/representativeness assessment involved a large number of experts (listed as co-authors of this study), with each expert working specifically on a dedicated list of elements, whose distribution was based on the individual expertise of each contributor.

2.2. LCI data quality/representativeness analysis: the Pedigree matrix

The quality and representativeness of LCI data were assessed using the Pedigree approach. In practice, this approach builds on the so-called Pedigree matrix, which provides a semi-quantitative rating framework to evaluate each individual data point within a given dataset, i.e. input and output flow. The Pedigree matrix used in this study is provided in Table S1.

Table S1: Pedigree matrix used in this study, slightly adapted from Muller et al. (2016)

Score	1	2	3	4	5
Reliability	Data based on i) measurements, or ii) physico-chemical calculations (e.g. stoichiometric calculations)	Data based on i) databases, or ii) models specific to the sector, or iii) experimental data	Data based on interpolation (i.e. manipulation) of multiple data (e.g. averages)	Data based on i) scientific and grey literature, or ii) estimations, or iii) experts judgment	Unknown data source
Completeness	Representative data from all sites relevant from the market considered.	Representative data from >75% of the sites relevant from the market considered.	Representative data from >50% of the sites relevant from the market considered.	Representative data from >25% of the sites relevant from the market considered.	Representative data from less than 5% of the sites relevant from the market considered. Or unknown.
Temporal correlation	Less than 3 years of difference to the time period of the dataset	Less than 6 years of difference to the time period of the dataset	Less than 10 years of difference to the time period of the dataset	Less than 15 years of difference to the time period of the dataset	Age of data unknown or more than 15 years of difference to the time period of the dataset
Geographical correlation	Data from area under study	Average data from larger area in which the area under study is included	Data from area with similar production routes	Data from area with slightly different production routes	Data from unknown or distinctly different area (North America instead of the Middle East, OECD-Europe instead of Russia)
Further technological correlation	Data from enterprises, processes and materials under study	Data from processes and materials under study (i.e. identical technology) but from different enterprises	Data from processes and materials under study but from different technology	Data on related processes or materials	Data on related processes on a laboratory scale or from different technology

3. LCI data verification and validation process

The overall process from LCI data compilation to processing and modelling included different verification and validation steps to ensure consistency and robustness in the final generated LCI datasets.

Given that a multitude of different experts contributed to this work on LCI data, the first key aspect was to ensure harmonisation in the:

- Compiled data and information from the original LCA studies;
- Rationale for data quality and representativeness rating through the Pedigree approach;
- Mapping of input and output flows with intermediate and elementary exchanges from the ecoinvent database (in line with the defined background modelling rules; see section 5), e.g. regarding energy or water flows;
- Calculations performed at individual input and output flow of the LCI datasets.

Moreover, this verification process also included a consistency check, to ensure the:

- Appropriate use of proxies in the absence of matching ecoinvent activities/flows (e.g. regarding chemicals);
- Correct implementation of multifunctionality-solving approaches in case of multi-outputs LCI datasets (e.g. application of subdivision, calculation of allocation factors).

Finally, a validation step was also implemented regarding the selection and exclusion of LCI datasets, based on a set of quantitative and qualitative criteria as well as expert-based judgement. This validation step was particularly crucial to ensure a sound and consistent datasets selection, a key milestone for the following steps of the study (i.e. generation of LCI models at production route and market levels).

4. Rationale for definition of the multifunctionality-solving approach

Multifunctionality issues arise whenever a product system delivers multiple functions/products. To address these issues, the ISO 14044 standard (ISO, 2006) provides a three-step hierarchy of approaches:

- i) Allocation should be avoided wherever possible by system subdivision or expansion (possibly including substitution).
- ii) If allocation is unavoidable, it should be first performed based on physical criteria (e.g. mass).
- iii) If no physical relationship can be used for allocation, it could then be based on other types of criteria (e.g. economic value).

In the case of metals co-production, Santero and Hendry (2016) recommend the use of system expansion as the first option for dealing with co-products when LCI data for single-output alternative routes are available, thereby eliminating the co-products from the product system. If system expansion is not applicable, they then recommend the use of allocation on a mass basis to deal with base metals and on an economic basis in the case of precious metals.

In addition to these recommendations, Lai et al. (2021) recommend privileging subdivision as the first option (similarly to the ISO standard), even if only partially applicable, with allocation as a complement to deal with the remaining multi-output unit processes. Alternatively, they advocate the use of system expansion when the co-products may be considered of limited importance with respect to the goal of the study (which implies they could be eliminated from the product system).

In line with these recommendations, (partial) subdivision was prioritised in this study wherever applicable. Economic allocation was then used as a second option to deal with metal co-products, as it enables better capturing discrepancies in economic values between metal products (Lai et al., 2021).

In this case, economic allocation was applied considering market prices of metal commodities averaged over 10 years in order to limit the effect of price volatility over time (as recommended by Santero and Hendry (2016)), provided that such averages are available (which may not be the case for all commodities). Price data were primarily drawn from Argus Media (2024¹), complemented with other sources when necessary. Shorter averages (e.g. three or five years) or proxies based on other commodities were used in case of data gaps for a given commodity. All prices used in this study are provided in the SI document n°2.

Finally, regarding non-metal co-products (e.g. sulfuric acid), system expansion (with substitution) was used if LCIs of single-output alternative production routes were available. If not, mass allocation was used, as recommended by Santero and Hendry (2016).

5. Background system modelling rules

The background system was modelled using the ecoinvent database in its 3.10 version, considering the cut-off system model.

To harmonise the LCI modelling as far as possible, a set of common background modelling rules was defined and applied to all selected datasets for developing LCI models, including generic guidelines as well as specific rules.

The generic rules aimed to provide general guidelines applicable to all LCI datasets. Firstly, flows from/to technosphere (intermediate exchanges) were to be associated with an ecoinvent activity, and flows from/to ecosphere (elementary exchanges) with an ecoinvent elementary flow. Secondly, transformation activities were to be used to model a given intermediate exchange when specific details as to geography and/or production pathway were available. Otherwise, market activities were to apply. Finally, regions of ecoinvent activities were to be selected in accordance with the location of the considered production route, with the following preference: i) country of the production route if available, ii) region/area including the country under focus (e.g. RER or RoW), and iii) world (GLO).

Specific rules were defined for mapping key intermediate exchanges (associated with ecoinvent activities) and elementary exchanges (associated with elementary flows): electricity, heat, fuels and gas, chemicals, raw materials, water, mineral resource, land use, waste disposal, emissions to air/water/soil. For several of these key exchanges, “specific” and “by default” options were defined in order to cover various possibilities depending on the level of information provided in each LCI dataset (which varies from one dataset to another).

Table S2 provides the full set of rules as defined in this study, including propositions of suitable ecoinvent activities and elementary flows for a number of possible input/output flows that may be reported in LCI datasets of metals and minerals production. It is to be noted that this set of specific rules aims to provide guidelines for many possible cases but does not intend to be comprehensive as it surely fails to cover a number of specific cases.

¹ <https://www.argusmedia.com/en>

Table S2: Specific background modelling rules defined in this study for mapping key intermediate and elementary flows

Flow category	Recommendations for background modelling	Examples of possible input or output flows in LCI	Examples of suitable ecoinvent activities/elementary flows
Electricity	By default, the "medium voltage" mix (i.e. market) of the country associated with the considered production route shall apply, if no specific details about the electricity source are provided by the original LCI developer.	Electricity	market for electricity, medium voltage
	If details are available, then the specific electricity source shall be modelled, in particular in the case of on-site electricity production.	Diesel (for electricity)	market for diesel, burned in diesel-electric generating set, 18.5kW
Heat	By default, the "heat, district or industrial, natural gas" shall apply, if no specific details about the heat source are provided by the original LCI developer.	Heat	market for heat, district or industrial, natural gas
	If details are available, then the specific heat source shall be modelled.	Heat from steam	market for heat, from steam, in chemical industry
Fuels and Gas	Diesel used in machinery or for electricity production shall be modelled with a diesel combustion activity.	Diesel	market for diesel, burned in building machine
	Any fuel or gas shall be associated with its corresponding activity.	Kerosene	market for kerosene
		Fuel oil	market for light fuel oil
		Natural gas	market for natural gas, high pressure
		Liquefied natural gas	market for natural gas, liquefied
		Air/Oxygen	market for compressed air, 800 kPa gauge
		Liquefied oxygen	market for oxygen, liquid
		Naphta	market for naphtha
		Steam	market for steam, in chemical industry
		Carbon dioxide	market for carbon dioxide, in chemical industry
		Liquefied carbon dioxide	market for carbon dioxide, liquid
	If no direct match for a given fuel can be found in the ecoinvent database, then the "light fuel oil" activity may be used as a proxy.	Gasoline	market for light fuel oil
		Mineral oil	market for light fuel oil

Chemicals	Any chemical shall be associated with its corresponding activity.	Hydrochloric acid (HCl)	market for hydrochloric acid, without water, in 30% solution state
		Nitric acid (HNO ₃)	market for nitric acid, without water, in 50% solution state
		Copper sulfate (CuSO ₄)	market for copper sulfate
		Sodium hydroxide (NaOH)	market for sodium hydroxide, without water, in 50% solution state
		Sodium carbonate (Na ₂ CO ₃)	market for soda ash, dense
		Sodium cyanide (NaCN)	market for sodium cyanide
		Sodium dichromate (Na ₂ Cr ₂ O ₇)	market for sodium dichromate
		Sulfuric acid (H ₂ SO ₄)	market for sulfuric acid
		Sulfur dioxide (SO ₂)	market for sulfur dioxide, liquid
		Iron sulfate (FeSO ₄)	market for iron sulfate
		Hydrogen peroxide (H ₂ O ₂)	market for hydrogen peroxide, without water, in 50% solution state
		Phosphoric acid (H ₃ PO ₄)	market for phosphoric acid, industrial grade, without water, in 85% solution state
		Methanol	market for methanol
		Ammonium nitrate	market for ammonium nitrate
		Triethylene glycol	market for triethylene glycol
		Sodium chloride (NaCl)	market for sodium chloride, brine solution
		Sodium sulfide (Na ₂ S)	market for sodium sulfide
		Active carbon	market for activated carbon, granular
		Calcium chloride (CaCl ₂)	market for calcium chloride
		Sodium metasilicate (Na ₂ SiO ₃)	market for sodium silicate, solid
		Ammonium chloride (NH ₄ Cl)	market for ammonium chloride
		Zinc sulphate (ZnSO ₄)	market for zinc monosulfate

	If no direct match for a given chemical can be found in the ecoinvent database, then a proxy containing the same active substance may be used.	Potassium amyl xanthate	market for potassium hydroxide
		Butanol	market for 1-butanol
		Calcium sulfate (CaSO ₄)	market for calcium chloride
		DiSodium tetraborate	market for borax, anhydrous, powder
	If no suitable proxy can be found in the ecoinvent database, then the "chemical, organic" or "chemical, inorganic" activities may be used by default, depending on the nature of the chemical.	Flocculant	market for chemical, inorganic
		Froth	market for chemical, organic
		Collector	market for chemical, organic
		Guar	market for chemical, organic
Raw materials	Regarding steel, the "steel, low-alloyed" activity shall be used by default, unless otherwise specified by the original LCI developer.	Steel	market for steel, low-alloyed
		Stainless steel	market for steel, chromium steel 18/8
	Any other raw material shall be associated with its corresponding activity or a proxy containing the same active material.	Explosives	market for explosive, tovox
		Concrete	market for concrete, normal strength
		Lime	market for lime, packed
		Cement	market for cement, unspecified
		Lead (II) oxide	market for lead
		High density polyethylene (HDPE)	market for polyethylene, high density, granulate
		Timber	market for glued laminated timber, average glue mix
		Rubber	market for synthetic rubber
		Polyurethane	market for polyurethane, rigid foam
		Glass fibre	market for glass fibre
		Pump	market for pump, 40W
		Conveyor	market for conveyor belt
		Polyacrylamide	market for polyacrylamide
		Polypropylene	market for polypropylene, granulate
		XPS	market for polystyrene, extruded
		Polycarboxylate	market for polycarboxylates, 40% active substance
		Ferrosilicon (FeSi)	market for ferrosilicon

		Quicklime / Calcium oxide (CaO)	market for quicklime, milled, packed
		Limestone / Calcium carbonate (CaCO ₃)	market for limestone, crushed, washed
		Manganese sulfate	market for manganese sulfate
		Hydrated lime / Calcium hydroxide (Ca(OH) ₂)	market for lime, hydrated, packed
Water	By default, any water input shall be associated with an elementary flow, with the water source if specified by the original LCI developer or with the "unspecified natural origin" flow otherwise.	Water	Water, unspecified natural origin
		Water from river	Water, river
		Cooling water	Water, cooling, unspecified natural origin
	If specified by the original LCI developer, the specific water production activity (e.g. tap water) shall be used for modelling the water supply.	Tap water	market for tap water
		Demineralized water	market for water, deionised
Mineral resource	Any input flow of mineral resource from the environment shall be associated with an elementary flow in the form of "Resource" with the subcompartment "in ground".	Iron	Iron (in ground)
Land use	Land use shall be modelled with different elementary flows: - An occupation flow, respectively associated with "mineral extraction site" in the case of mining and "industrial area" in the case of refining.	Land use	Occupation, mineral extraction site
			Occupation, industrial area
	- A "transformation from" flow, associated with "unspecified" unless otherwise specified by the original LCI developer.	Land use	Transformation, from unspecified
	- A "transformation to" flow, respectively associated with "mineral extraction site" in the case of mining and "industrial area" in the case of refining.	Land use	Transformation, to mineral extraction site
			Transformation, to industrial area
Waste disposal	Any waste shall be modelled with a disposal/market activity, corresponding as much as possible to the type of waste specified in the LCI.	Steel	treatment of scrap steel, inert material landfill
		Slags	market for slag landfill
		Nickel slags	market for nickel smelter slag
		Overburden	market for non-sulfidic overburden, off-site

	If no direct match for a given type of waste can be found, then the "inert waste, for final disposal" activity may be used as a proxy.	Mixed waste	market for inert waste, for final disposal
	Regarding tailings, the latter shall be modelled with a specific tailings disposal activity (either sulfidic or non-sulfidic). In the case of sulfidic tailings, the activities corresponding to the metals and countries/regions of the production route shall be used (if available). Otherwise, the "generic" tailings disposal activity shall be used by default.	Tailings	market for sulfidic tailings, generic
Emissions to air	Any dust emission shall be modelled as a PM emission, with the corresponding granulometry. If unknown, the "PM < 10 um flow" shall apply.	Dust - PM10	Particulate Matter, > 2.5 um and < 10 um
		Dust - PM2.5	Particulate Matter, < 2.5 um
		Dust > 10 microns	Particulate Matter, > 10 um
	Any emissions to the environment shall be associated with an elementary flow corresponding to the emitted substance or its main active substance if no direct match is available.	CO ₂	Carbon dioxide
		NH ₃	Ammonia
		CO	Carbon monoxide
		SO _x	Sulfur oxides
		NO _x	Nitrogen oxides
		Volatile Organic Compounds (VOC)	VOC, volatile organic compounds
Emissions to water	Any emissions to the environment shall be associated with an elementary flow corresponding to the emitted substance or its main active substance if no direct match is available.	NO ₂	Nitrogen
		N total	Nitrogen
		Nitrates	Nitrate
Emissions to soil	Any emissions to the environment shall be associated with an elementary flow corresponding to the emitted substance or its main active substance if no direct match is available.	Diesel	Oils, unspecified

6. Market shares estimation approach: parameters, data sources and assumptions

The approach for estimating market shares of each individual production route considered in this study was developed considering a set of several parameters respectively related to geographical, geological, technological and material characteristics. These parameters were quantified based on different data sources, with potentially different levels of granularity in the available data. In the case of data gaps, a number of assumptions was also considered as a way to address the gaps.

The full set of parameters, as well as associated assumptions and data sources, is provided in Table S3.

Table S3: List of parameters considered in the market shares estimation approach, including descriptions, assumptions and data sources

Category	Parameter	Description of the parameter and possible assumptions	Main associated data sources
Geography	Processing country share $\%_{PC_{i,j}}$	<u>Description</u> Reflects the contribution from the processing country j to the global production of the element i . Here processing is understood as covering purification and refining stages.	• EC (2023) • WMD (2023)
	Site-specific share $\%_{Site_{i,j}}$	<u>Description</u> Reflects the contribution from the specific production site to the national production (country j) of the element i (e.g. Salar de Atacama in Chile).	• S&P Global (2023) • Company websites • Literature
	Provincial share $\%_{Prov_{i,j}}$	<u>Description</u> Reflects the contribution from the specific province to the national production (country j) of the element i (e.g. Jiangxi province in China).	• Literature
Geology and technology	Geological share $\%_{Geol_{i,j}}$	<u>Description</u> Reflects the share of production (potentially at site, national or global scale) from the specific geological deposit (e.g. lateritic or sulfidic Ni ores) <u>Possible assumption</u> In the absence of detailed data, global data related to deposit distribution are assumed to apply similarly at country-level.	• Literature • Expert-based assumptions

	Process share $\%_{Proc_{i,j}}$	<u>Description</u> Reflects the share of production (potentially at national or global scale) from the specific process chain or technology (e.g. pyrometallurgy, hydrometallurgy). <u>Possible assumption</u> - In the absence of detailed data, global data related to process distribution are assumed to apply similarly at country-level. - In the absence of detailed data, if several processes associated with a given element are considered widespread (according to literature or expert-based judgement), an equal distribution of processes/technologies (50/50) may be assumed.	• Literature • Expert-based assumptions • International study groups
	EoL RIR share $\%_{RIR_{i,j}}$	<u>Description</u> Reflects the contribution from secondary supply to the global production of the element i <u>Possible assumption</u> In the absence of country-level data related to secondary production, primary production shares per country are assumed to apply similarly to secondary production (assuming that secondary and primary material streams are mixed within processing plants).	• Talens Peiro et al. (2018) • UNEP (2011) • Literature
	Waste stream share $\%_{W_{i,j}}$	<u>Description</u> Reflects the share of secondary production from old scraps originating from the specific waste stream (e.g. construction & demolition wastes). <u>Possible assumption</u> In the absence of detailed data, global data related to waste stream distribution are assumed to apply similarly at country-level.	• Literature • Expert-based assumptions

Material	Commodity share $\%_{Comm_{i,j}}$	<u>Description</u> Reflecting the share of production (potentially at site, national or global scale) under the form of the considered commodity (e.g. NiSO ₄ , Ni class 1).	• SCREEN (2023) • Literature • Expert-based assumptions
		<u>Possible assumption</u> - In the absence of detailed data, global data related to commodities distribution are assumed to apply similarly at country-level. - The commodity share only applies to primary production routes, assuming a more limited variety of commodities as direct outputs from secondary routes.	

7. Final selection of LCI datasets and associated data sources

The final selection of LCI datasets builds on 81 different sources (i.e. LCA studies) out of the 308 initially identified. While scientific papers constitute the main data sources, a few data were directly drawn from publications by industrial associations, where the LCI datasets were developed from industry-based data. Notably, most LCI datasets of Al were directly drawn from such associations, namely the International Aluminium Association (2022), The Aluminum Association (2022), and European Aluminium (2018). Similarly, regarding PGM, the two LCI datasets originate from the International PGM association, while for Ni one LCI dataset was drawn from the Nickel Institute (2021). Finally, from a temporal perspective, the large majority (54) of the 81 studies here considered as data sources were issued after 2018, of which another large share was released after 2021 (32). On the other hand, five references dating back to 2010 or prior were also considered in this study, in the absence of more recent data (Figure S2).

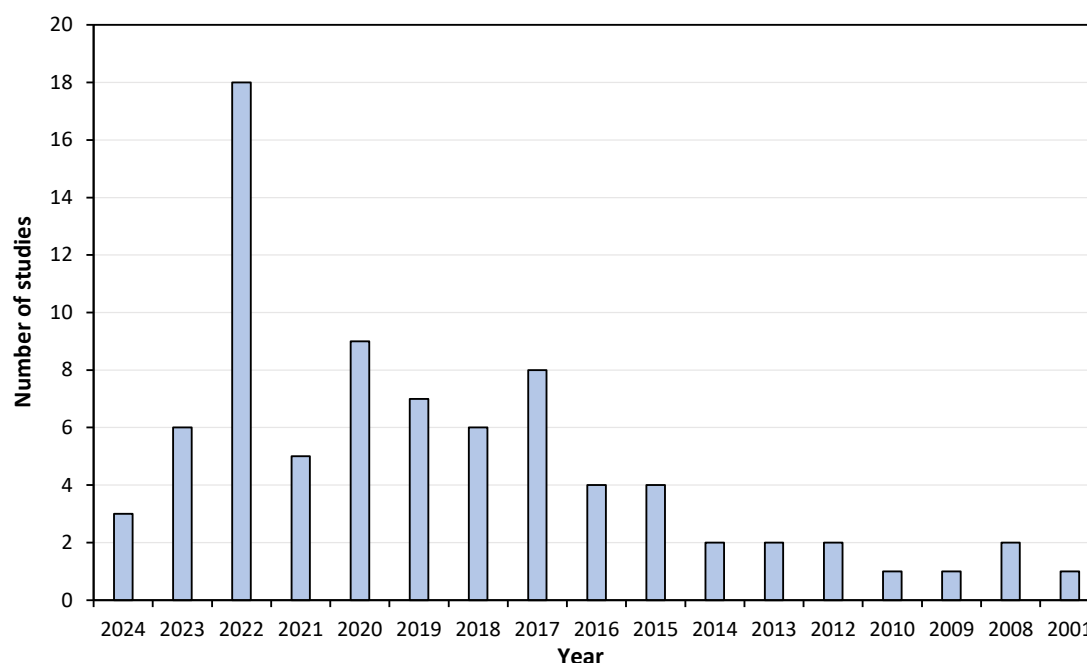


Figure S2: Reference years of the LCA studies, sources of the selected LCI datasets

References

- EC (European Commission), 2023. Study on the Critical Raw Materials for the EU 2023 - Final report.
- European Aluminium, 2018. Environmental profile report - Life-Cycle inventory data for aluminium production and transformation processes in Europe. Available at: https://european-aluminium.eu/wp-content/uploads/2023/01/European-Aluminium_Environmental-Profile-Report-2018_full-version.pdf.
- International Aluminium Association, 2022. Life cycle inventory data and environmental metrics for the primary aluminium industry. Available at: <https://international-aluminium.org/resource/2019-life-cycle-inventory-lci-data-and-environmental-metrics/>.
- ISO, 2006. 14044: Environmental management – Life Cycle Assessment – Requirements and guidelines. International Organisation for Standardisation.
- Lai, F., Laurent, F., Beylot, A., Villeneuve, J., 2021. Solving multifunctionality in the carbon footprint assessment of primary metals production: Comparison of different approaches. *Miner. Eng.* 170, 107053. <https://doi.org/10.1016/j.mineng.2021.107053>
- Muller, S., Lesage, P., Samson, R., 2016. Giving a scientific basis for uncertainty factors used in global life cycle inventory databases: an algorithm to update factors using new information. *Int. J. Life Cycle Assess.* 21, 1185–1196. <https://doi.org/10.1007/s11367-016-1098-5>
- Nickel Institute, 2021. Life cycle assessment report: nickel sulphate hexahydrate.
- Santero, N., Hendry, J., 2016. Harmonization of LCA methodologies for the metal and mining industry. *Int. J. Life Cycle Assess.* 21, 1543–1553. <https://doi.org/10.1007/s11367-015-1022-4>
- SCRREEN, 2023. SCRREEN raw materials factsheets. Available at: <https://screen.eu/crms-2023/>.
- S&P Global, 2023. Available at: <https://www.capitaliq.spglobal.com/web/client?auth=inherit#dashboard/metalsAndMining>.
- Talens Peiro, L., Nuss, P., Mathieux, F., Blengini, G., 2018. Towards Recycling Indicators based on EU flows and Raw Materials System Analysis data, EUR 29435 EN, Publications Office of the European Union, Luxembourg, 2018, ISBN 978-92-79-97247-8 (online), doi:10.2760/092885 (online), JRC112720.
- The Aluminum Association, 2022. The Environmental Footprint of Semi-Fabricated Aluminum Products in North America. Available at: https://www.aluminum.org/sites/default/files/2022-01/2022_Semi-Fab_LCA_Report.pdf.
- UNEP (United Nations Environment Programme), 2011. Recycling Rates of Metals - A Status Report, A Report of the Working Group on the Global Metal Flows to the International Resource Panel.
- WMD, 2023. World Mining Data 2023.